AOC-IDS: Autonomous Online Framework with Contrastive Learning for Intrusion Detection

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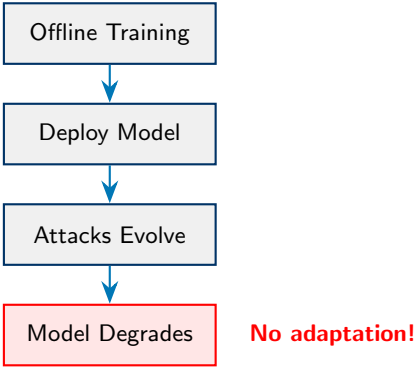
Outline

1. Background & Motivation
2. AOC-IDS Framework
3. Our Novel Contributions
4. Summary

The Problem with Traditional IDS

Challenges in real-world intrusion detection:

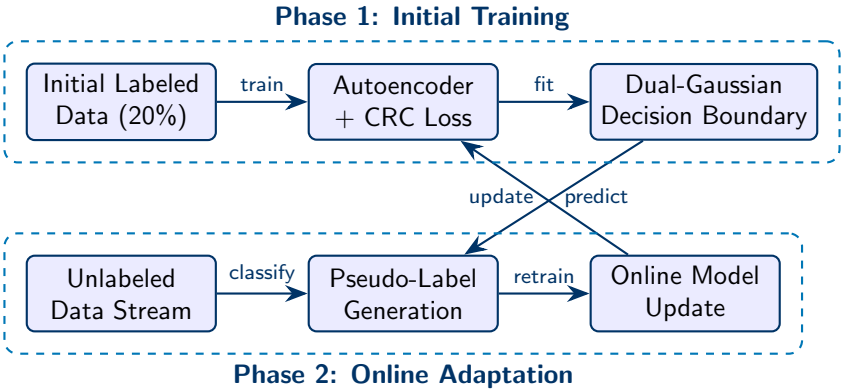
- Traditional IDS relies on **offline, static training**
- Attack strategies **evolve continuously**
- Manual labeling of network traffic is **expensive** and **slow**
- IoT environments generate **massive, dynamic** data streams
- Models degrade over time without retraining (**concept drift**)



Key Question

How can an IDS *autonomously* adapt to new traffic patterns *without* human labeling?

AOC-IDS: System Overview



Core Component: Autoencoder + CRC Loss

Autoencoder Architecture

- Encoder: $d \to d/2 \to d/4$ (bottleneck)
- Decoder: $d/4 \to d/2 \to d$ (reconstruction)
- Outputs **two** representation spaces:
  - Encoder embeddings (compressed features)
  - Decoder reconstructions

Total Loss:

$$\mathcal{L} = \mathcal{L}_{\text{CRC}}^{\text{encoder}} + \mathcal{L}_{\text{CRC}}^{\text{decoder}}$$

Cluster Repelling Contrastive (CRC) Loss

- Based on InfoNCE, but with **two traversals** of the abnormal set
- Pulls normal samples **together**
- Pushes normal and abnormal samples **apart**

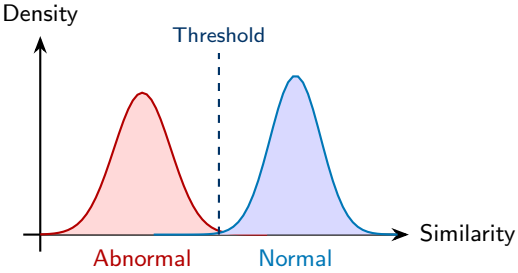
$$\mathcal{L}_{\text{CRC}} = -\frac{1}{N_n} \sum_{i \in \mathcal{N}} \log \frac{e^{s_{ij}/\tau}}{e^{s_{ij}/\tau} + \sum_{k \in \mathcal{A}} e^{s_{ik}/\tau}}$$

where s_{ij} = cosine similarity, τ = temperature, $\mathcal{N}$ = normal, $\mathcal{A}$ = abnormal

Detection: Dual-Gaussian Decision Boundary

How detection works:

1. Compute **normal template** = mean embedding of all normal training samples
2. For each sample, compute **cosine similarity** to normal template
3. Fit **two Gaussians** (via MLE) to the similarity distribution:
   - $\mathcal{N}(\mu_1, \sigma_1)$ — normal cluster (high similarity)
   - $\mathcal{N}(\mu_2, \sigma_2)$ — abnormal cluster (low similarity)
4. Classify: if $p(\text{abnormal} \mid x) > p(\text{normal} \mid x)$ $\Rightarrow$ **attack**

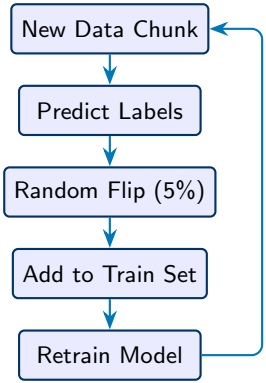


Final vote: encoder and decoder predictions combined by confidence ($|p_1 - p_2|$).

Online Pseudo-Labeling Loop

How online adaptation works:

1. Start with a small labeled set (20% of training data)
2. Train the AE + CRC model on this initial set
3. Receive a **chunk** of unlabeled data (size = `sample_interval`)
4. Use Gaussian decision boundary to **predict pseudo-labels**
5. **Flip** a small fraction of labels (5%) for robustness
6. Add pseudo-labeled data to training set
7. **Retrain** model for `epoch_1` epochs
8. Repeat until all data is consumed



Experimental Results

Datasets:

- **NSL-KDD** — 121 features, binary labels
- **UNSW-NB15** — 196 features, binary labels
- Preprocessed: normalized continuous, one-hot categorical

Setup:

- 20% initial labeled data
- 80% arrives as unlabeled stream
- 5 random seeds, averaged results
- Evaluated: Accuracy, Precision, Recall, F1

Key findings from the paper:

- AOC-IDS **outperforms** state-of-the-art on both datasets
- Achieves strong results with **only 20%** labeled data
- Online adaptation **improves** performance over the static baseline
- CRC loss is **more effective** than standard contrastive losses (InfoNCE)

Significance

First framework combining contrastive learning with autonomous online adaptation for IDS.

Novelty 1: GMM-Based Adaptive Decision Boundary

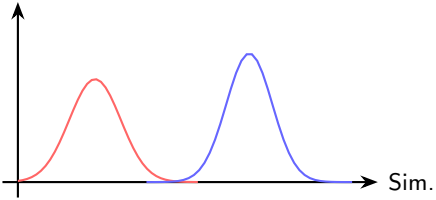
Problem with original approach:

- Fixed **2-Gaussian** assumption
- Multiple attack types (DDoS, Probe, R2L...) form **different clusters**
- Nelder-Mead optimizer is **unconstrained** — can produce negative σ

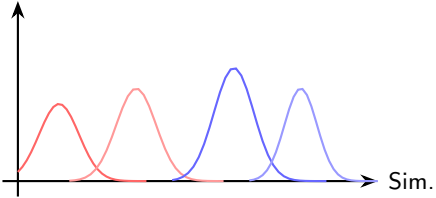
Our improvement:

- Replace with **Gaussian Mixture Model (GMM)**
- Auto-select components ($k = 2$-5) via **BIC**
- Label components as normal/abnormal using **training label majority**

Original: 2 Gaussians



Novel: GMM (BIC-selected)



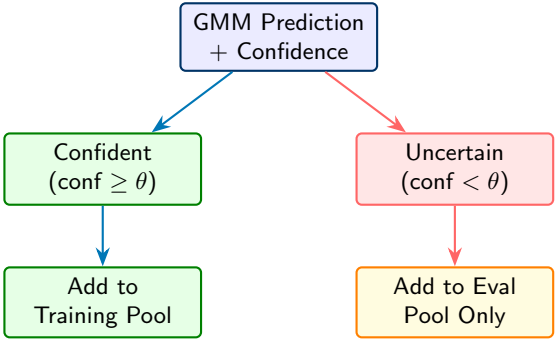
Novelty 2: Confidence-Aware Pseudo-Label Filtering

Problem with original approach:

- **All** pseudo-labels added to training — including wrong ones
- Random flipping (5%) is **blind**: flips correct labels too
- Noisy labels **accumulate** over online rounds
- Model performance degrades with error propagation

Our improvement:

- Compute **confidence** = $|P(\text{normal}) - P(\text{abnormal})|$ from GMM posteriors
- Only add samples with confidence $> \theta$



Result: Cleaner training data $\Rightarrow$ better model $\Rightarrow$ better future predictions (virtuous cycle)

Novelty 3: Dynamic Temperature Scheduling

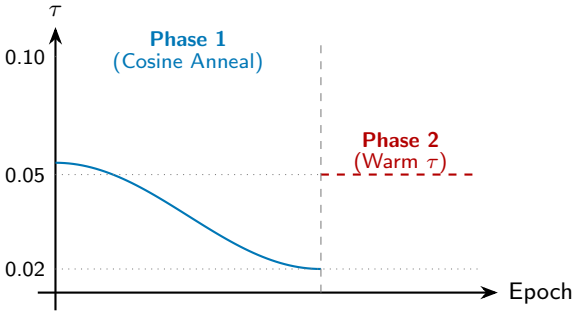
Problem with original approach:

- Fixed temperature $\tau = 0.02$ throughout training
- Too aggressive early on (embeddings still forming)
- Same τ for initial and online phases, but they have **different needs**

Our improvement — Cosine Annealing:

$$\tau(t) = \tau_{\min} + \frac{\tau_{\max} - \tau_{\min}}{2}\left(1 + \cos\frac{\pi\, t}{T}\right)$$

- **Phase 1** (initial): τ anneals from $0.1 \rightarrow 0.02$



High τ = soft separation (exploration)

Low τ = hard separation (exploitation)

Summary of Contributions

Aspect	Original AOC-IDS	Our Improvement
Decision Boundary	Fixed 2-Gaussian MLE	GMM with BIC selection
Pseudo-Label Quality	Random flip (5%)	Confidence-based filtering
CRC Temperature	Fixed $\tau = 0.02$	Cosine annealing + warm online τ

Why These Improvements Matter

- **GMM**: Captures multi-modal attack distributions $\Rightarrow$ better boundaries
- **Confidence filtering**: Breaks error propagation cycle $\Rightarrow$ cleaner online learning
- **Temperature scheduling**: Matches training dynamics $\Rightarrow$ stable convergence

Thank You!

Code: `novel_online_training.py` — `novel_utils.py`